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ASSESSING THE RISK FACTORS OF UNDER-FIVE CHILD MORTALITY: A STUDY BASED ON BDHS 2022

Abu Sufiun Rhyme

Institute of Statistical Research and Training (ISRT), University of Dhaka, Bangladesh

Introduction

Under-five child mortality is a key indicator of child health and health system performance. Although Bangladesh has achieved a substantial decline in under-five mortality from 143.8 per 1,000 live births in 1990 to 31 per 1,000 live births in 2022 preventable child deaths and regional disparities remain. Factors such as parental education, birth characteristics, healthcare access, and place of residence continue to influence child survival.

Objective of the study

This study aims to estimate the prevalence of under-five child mortality in Bangladesh and identify its associated individual- and regional-level determinants using nationally representative BDHS 2022 data, to support evidence-based policies for achieving SDG 3.2.

Data and Methodology

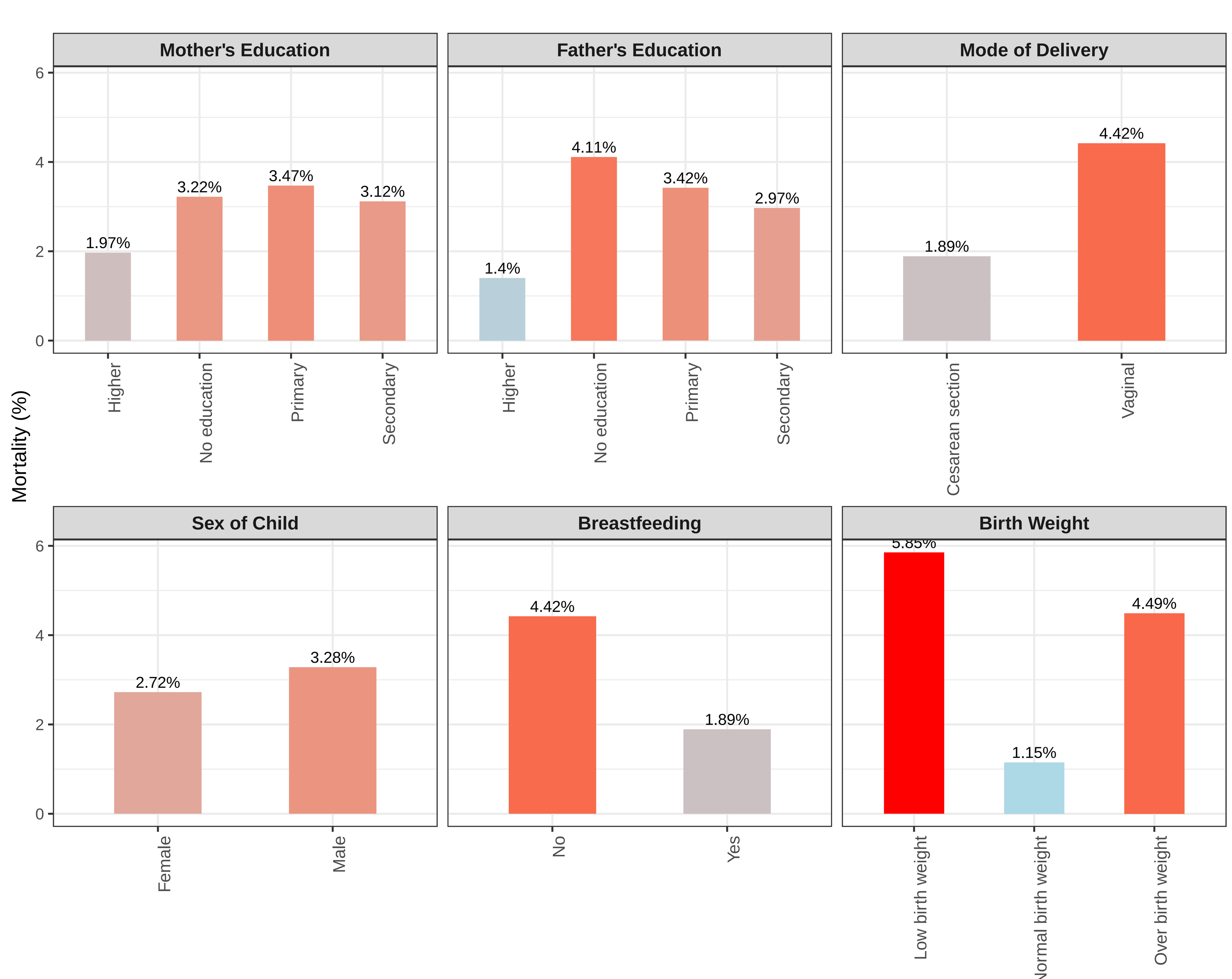
This study used secondary data from the Bangladesh Demographic and Health Survey (BDHS) 2022, a nationally representative survey conducted using a stratified two-stage cluster sampling design across eight administrative divisions. Data from 30,358 ever-married women aged 15–49 years were included in the analysis.

The outcome variable was under-five child mortality, coded as 1 if a child died at or before age five and 0 otherwise. Explanatory variables were selected based on prior literature and categorized into child-level (sex, birth order, birth weight, breastfeeding, delivery characteristics), parental (education, maternal age at birth, antenatal care, media exposure), and household-level factors (wealth index, residence, and division).

Descriptive statistics and Chi-square tests were used to examine unadjusted associations. A multilevel binary logistic regression model was applied to account for clustering and regional variation. Results were presented as adjusted odds ratios with a 5% level of significance. Model performance was assessed using the Receiver Operating Characteristic (ROC) curve.

Results

Figure 1: Child Mortality by Various Factors



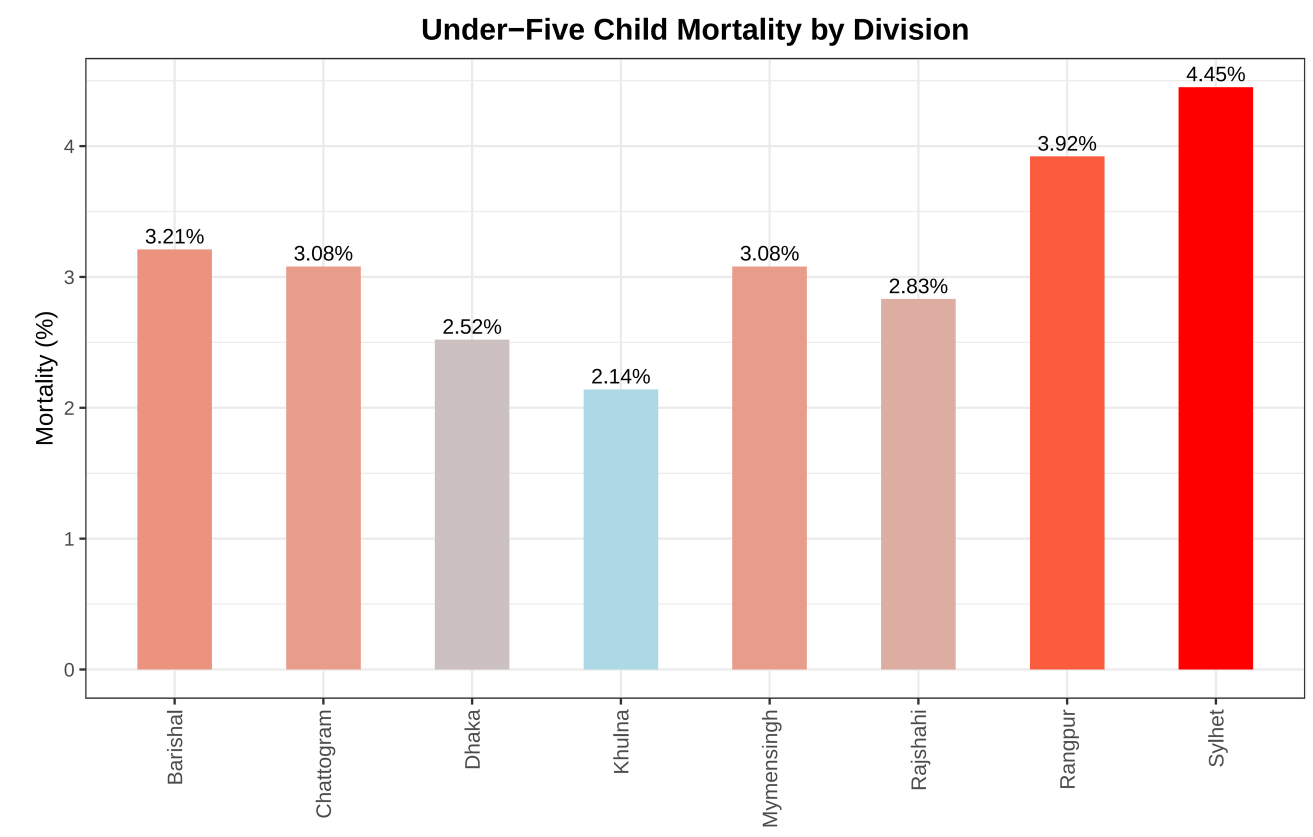


Table 1: Multilevel logistic regression results: significant determinants of under-five child mortality ($p < 0.05$)

Variable	Category	Adjusted Odds Ratio (OR)	95% CI
Division	Mymensingh (Ref.)	-	-
	Barishal	0.444	(0.104, 1.904)
	Chittagong	0.116	(0.027, 0.506)
	Dhaka	0.168	(0.040, 0.699)
	Khulna	0.451	(0.113, 1.798)
	Rajshahi	0.359	(0.078, 1.662)
	Rangpur	0.968	(0.264, 3.551)
	Sylhet	0.233	(0.051, 0.997)
Mother's education	No education (Ref.)	-	-
	Primary	0.975	(0.591, 1.612)
	Secondary	0.899	(0.559, 1.447)
	Higher	0.957	(0.302, 0.945)
Father's education	No education (Ref.)	-	-
	Primary	4.552	(1.192, 17.386)
	Secondary	2.020	(0.500, 8.157)
	Higher	0.957	(0.158, 5.811)
Sex of child	Female (Ref.)	-	-
	Male	2.117	(1.016, 4.414)
Birth weight	Low (Ref.)	-	-
	Normal	0.113	(0.051, 0.250)
	Over	0.428	(0.134, 1.368)
Breastfeeding	No (Ref.)	-	-
	Yes	0.052	(0.023, 0.113)
Mode of delivery	Vaginal (Ref.)	-	-
	Caesarean	0.307	(0.125, 0.746)

- **Intraclass Correlation Coefficient (ICC):** 0.122 (95% CI: 0.092, 0.207), indicating that 12.2% of the variation in under-five child mortality is attributed to cluster-level differences.
- **Receiver Operating Characteristic (ROC) curve AUC:** 0.9109, suggesting high discrimination ability of the fitted multilevel logistic regression model.

Discussion & Conclusions

Key Findings

Higher parental education reduces under-five mortality. Male sex, low birth weight, and lack of breastfeeding increase risk. Caesarean delivery lowers mortality. Mortality varies by region, highest in Mymensingh.

Policy Implications

Promote parental education, strengthen breastfeeding and safe delivery programs, and target high-risk regions with improved health services.

Limitations

Cross-sectional data limits causal inference. Some relevant risk factors were unavailable.